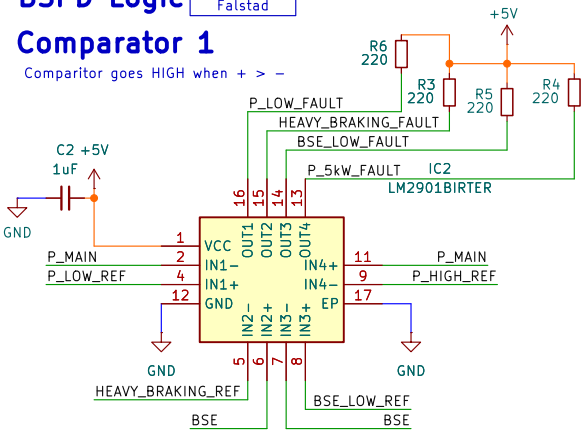
BSPD Logic

Logic in Falstad

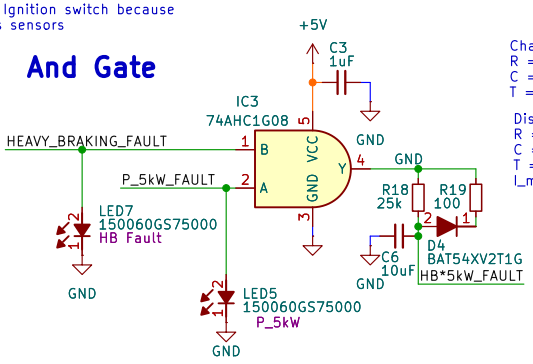
Logic needs to be powered off of Ignition switch because they get powered at same time as sensors

Comparator 1

Comparitor goes HIGH when + > −



And Gate

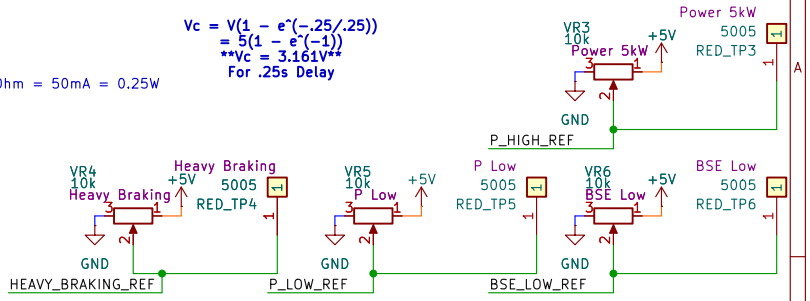


Charging Time
R = 20,000 Ohm
C = 0.00001 F
T = R*C = 0.25s

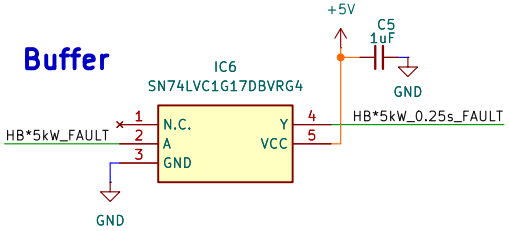
Discharging Time
R = 100 Ohm
C = 0.00001 F
T = R*C = 0.001s
I_max = 5V/100 Ohm = 50mA = 0.25W

Variable Resistors

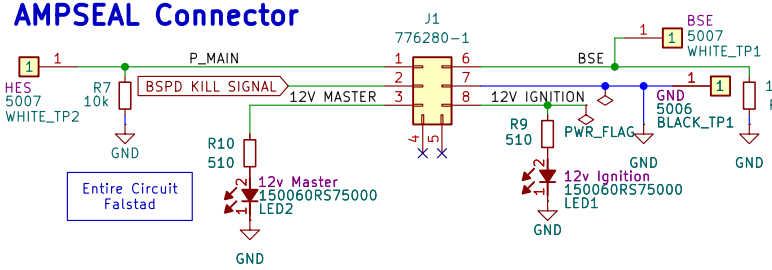
Vc = V(1 − e^(−.25/.25))
= 5(1 − e^(−1))
Vc = 3.161V
For .25s Delay



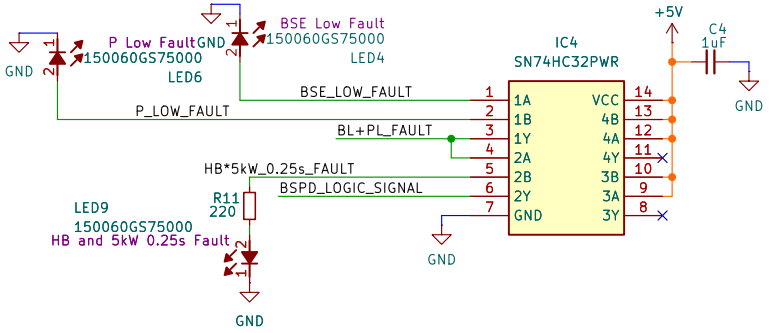
Buffer



AMPSEAL Connector

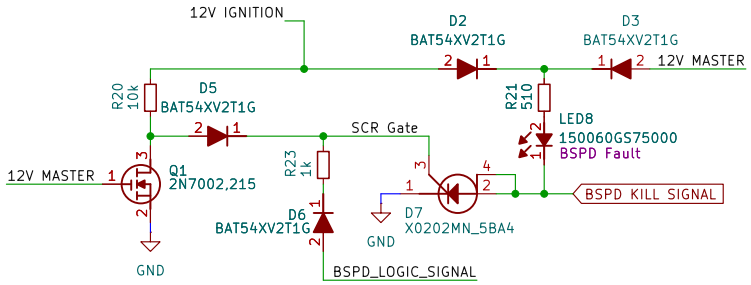


Or Gate

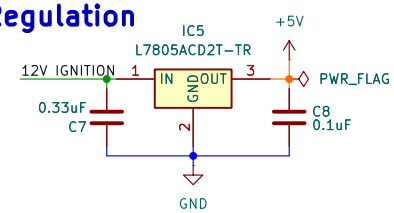


Latch

Latch in Falstad



5V Regulation



Sheet: /	
File: BSPD_V1.2_EV.kicad_sch	
Title: BSPD_V1.2_EV	
Size: A4	Date:
KiCad E.D.A. 10.0.7	Rev: Id: 1/1